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Analysis of Financial Budgeting Using Computational Intelligence Approaches: its need and importance in reference to the state of Uttarakhand

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***Abstract* - Financial budgeting plays a crucial role in effective resource allocation and economic stability, particularly in regions with unique geographical and economic constraints, such as Uttarakhand. Traditional budgeting methodologies often fall short in handling the complexities of modern financial ecosystems. This paper presents a comprehensive overview on the application of computational intelligence techniques, including artificial intelligence (AI), machine learning (ML), and big data analytics, in financial budgeting and resource optimization. The review systematically categorizes existing studies, identifies research gaps, and highlights the potential of AI-driven approaches in improving financial decision-making and policy formulation in Uttarakhand. Furthermore, it discusses challenges, ethical considerations, and real-world applications of these technologies in financial forecasting. The study underscores the necessity of integrating computational intelligence for dynamic and data-driven financial planning, aiming to enhance economic growth and resource efficiency in the region.**

Keywords - Financial Budgeting, Artificial Intelligence, Computational Intelligence, Machine Learning, Big Data Analytics, Public Finance, Uttarakhand, Resource Optimization

I. INTRODUCTION

Financial budgeting can be defined as a tool for management of any business or economic structure by means of flow of cash, assets, expenditure or income. In case of government structure, financial budgeting can be defined as a tool used for preparation of statement of expected revenues as well as expenditures for a financial year. The statement of revenues and expenditure is prepared in accordance with the objectives of government policies while keeping in account the legislative scrutiny and approval. Thus, it can be worth mentioning that financial budgeting plays a critical role for understanding the economic structure as well as fiscal health of a country or a state.

Financial budgeting covers all the revenues generated by a state or a country by means of taxes, grants or income from public sector undertakings as well as expenditure encompassing expenses on infrastructure, education, social work etc. Due to this it becomes a crucial factor in a state or country which needs to be analyzed for efficient planning, management, estimation as well as allocation of budgets and resources to different departments for achieving specific goals within specified time limits.

With respect to state of Uttarakhand, which is a hilly state with limited resources, financial budgeting is one of the major aspects that plays a pivotal role for ensuring proper financial management, effective allocation of resources as well as achieving sustainable economic growth. Additionally, financial budgeting helps in evaluating the financial position of Uttarakhand at any specified time period which will help in strategic decision-making. Uttarakhand, where economic activities are generated by factors like tourism, agriculture, and industrial development, financial planning must be backed by accurate and real-time data. Proper financial data analysis allows policymakers to track revenue trends, assess expenditure efficiency, and optimize resource allocation.

With advancement in Information Technology (IT) in line with artificial intelligence, machine learning, big data analytics, financial budgeting in Uttarakhand will be helpful in enhancing financial governance, improving transparency and enabling data-driven decision making. The structured financial data helps in enabling the government to ensure financial

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sustainability, promoting economic growth as well as improving overall standard of life of people of Uttarakhand.

Currently, the departments of Government of Uttarakhand (GoUK) follows the budget manual of state government for following of budget calendar. It has been observed that planning and budget preparation is done generally on an incremental basis. The approach for budgeting should be based on long-term view which is lacking. The schemes proposed under budget should be prioritized without this approach leads to distortion in coverage. One of the main factors affecting planning and budget cycle are cost and time overrun. Hence, resources assessments and allocation for various sectors should be based on priority indicated by state government on annual basis, these priorities may change affecting quality and outcome of expenditure. This justifies the need not only for adequate, accurate and effective financial data but also efficient data collection, effective data analyzation and accurate data visualization techniques for budget making in a resource constraint state like Uttarakhand.

The available information of the budget making and financial planning of the government of Uttarakhand shows serious lack of use of modern computational intelligence approaches. In the context of financial budgeting in Uttarakhand, this paper explores application of computational intelligence in order to understand how modern technological advancements can enhance financial planning and decision-making. Given the rapid advancements in AI and big data analytics, it is imperative to evaluate their role in optimizing financial management, improving forecasting accuracy, and ensuring efficient resource allocation.

II. METHODOLOGY:

For Selection of literature, snowball approach technique is used. The Snowball Approach technique was largely adopted for a comprehensive study about the topic. Initial source selection was done by collecting research papers, books, and reports from reputable databases such as IEEE, Springer, Elsevier, and Shodh Ganga. After reviewing the initial papers, citations and references within those studies were examined and relevant works on financial planning, AI, mathematical modeling, and government-level budgeting were identified and included in the review.

Fig. 1: Literature Review Methodology

As more studies were collected, strict inclusion and exclusion criteria was applied for filtering and refinement. Studies outside financial planning, resource optimization, or AI applications were discarded. Non-peer-reviewed and unreliable sources were also excluded to maintain academic integrity. The accumulated literature is now synthesized to form a comprehensive, up-to-date perspective on financial planning and budgeting. The iterative nature of this approach allows for a progressive expansion of sources, ensuring that the study remains comprehensive, relevant, and well-grounded in existing research

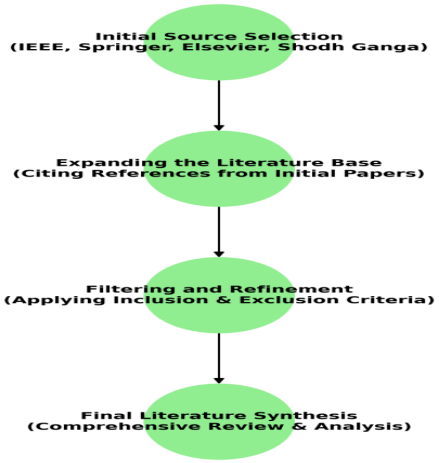

Fig. 2: Process of Literature synthesis

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III. Review Analysis of Financial Budgeting

The papers that use different computational approaches for financial budgeting were explored and further classified into three categories based on technologies used for financial budgeting namely traditional mathematical approach, artificial intelligence approach and big data approach.

Fig. 3: Classification of Financial budgeting methods

Traditional Mathematical Approach for Financial Management:

D. Acharya et. al., 2009, perform their study on net state domestic product (PCNSDP) and per capita credit (PCC) for 14 Indian states. These study performed from year 1981-2002 for states of Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Maharashtra, Rajasthan, Gujrat, Bihar, West Bengal, Assam, Andhra Pradesh, Kerala, Tamil Nadu. For analysis of relation between economic growth and financial development Pedroni Panel Co integration Test and Fully Modified Ordinary Least Squares (FMOLS) were used along with Vector Auto Regression (VAR) was employed to address endogeneity. They classify data into backward states, which they termed as BIMAARU and developed states. The states which have weaker credit output were categorized as BIMAARU states which need regulatory intervention and need for addressing structural issues in real sector. The study shows that the financial development will positively help in influencing economic growth for BIMAARU as well as non-BIMAARU states.

Zhang J and Su M., proposed a framework for evaluation of science and technology (sci-tech) financial ecosystems for various countries which helps in integrating finance with technological innovation. Overall evaluation is done on enterprise financing capabilities, enterprise innovation capability, fintech development, government service capacity, degree of financial openness and index system design. The study was conducted by collection of data from 2010 to 2019 followed by processing of data along with assigning the weights and finally applying structured methodology for a nuanced comparison of countries' sci-tech financial ecosystems, providing insights into their respective strengths and demarcation of areas that need development.

Chakrabarti D. et.al., 2018 make use of theoretical model and empirical analysis for examining the directed credit policies on financial development, financial liberalization, firm financing and economic growth. The study conducted on 932 Indian manufacturing firm from year 1990-2016. The research make use of panel data analysis, ARDL modeling and structure break test which helps in indicating that liberalization shifted firms towards equity financing. The research reveals that credit program can support financial development and economic growth under specific conditions.

Artificial Intelligence Approach for Financial Management:

Broby D., 2022 presents a survey paper where financial analytics is explained using computational techniques. This paper uses methods of classification, regression, clustering, and association along with time series model. This specifies that external data can be used to forecast the economy and corporate earnings. This analytics is performed using predictive models which analyze external data using historic time series to forecast the returns. It is mentioned that the descriptive data is used in finance for quantifying the relationship, thus helping in prediction of risks.

Chakri P. et. al., 2023 uses data analytics for financial accounting which include statement of income, balance sheets and cash flow statements by making use AI based approach by making use of AI techniques like K-nearest neighbor, linear regression, decision tree model and support vector regressor. It was found that out of these four, decision tree model results in optimized performance with maximum depth of nine by means of grid search. The indicators used for evaluation of performance are debt-to-equity, EPS, Inventory turnover and net margin.

Yang R. et. al in year 2019 forecast volatility of financial market using big data analysis and AI based support vector machine (SVM) model. In this approach, two prediction model were used which are based on neural network as recursion dynamic network prediction and window static network prediction. Prediction of financial data is done using SVM. The analysis was done on Shanghai stock market making use of non-linear prediction. A non-linear CARRX model based on LSSVM is proposed which provides better accuracy whereas SVM helps for forecasting of financial and regression forecasting model of least squares support which is constructed accordingly.

Akanni A, 2024 explains use of AI in financing and budgeting by taking case studies. In this approach, machine learning is used for analysis of real-time data along with history of data which help for creation of models used for prediction of financial scenarios for future with precision. They uses neural network for

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modeling non-linear relationship, complex model and more advanced by considering multitude of variables and interaction among them. For foreseeing the potential financial outcomes, predictive analysis is used and adjusting their strategies thus helps in fostering proactive rather than reactive approach for financial management. This AI-driven quantitative analysis budgeting models helps in providing valuable insight for AI for enhancing dynamic budgeting and forecasting.

Vineet Jain and Parth Kulkarni, 2023 make use of AI techniques to enhanced forecasting in financial and budgeting strategies. Automatic resource allocation and real-time adjustment is done using AI techniques. Thus, AI helps financial planners to concentrate on strategic analysis, aligning financial strategies seamlessly with organizational objectives.

Wasserbacher H. et. al., 2022 uses machine learning for forecasting, planning and analysis of financial data. In this paper, they make use of simulation approach to explain how machine learning can be used for forecasting and planning. They make use of Naïve application of machine learning for simulation purpose. The simulation design helps in projecting the settings, data types and questions that a planning and analysis department of a company could face. The main focus of this research is on accuracy of sales forecasting and effectiveness of promotional activities for sale.

Chen and yaping, 2021 explain uses of artificial intelligence in financial and accounting management systems. This is explained using intelligent accounting model which helps in enhancing efficiency, financial process optimization along with support for real-time decision making. This paper proposed three AI driven algorithm as Earliest Deadline First (EDF), Highest Value First (HVF), and Highest Value Density First (HVDF) which helps in enhancing the financial management. The paper also proposes shared accounting service model meant for centralized operations and real-time analysis.

Wall L., 2018 explain the regulatory implication of AI in finance. In this research the focus is on impact of machine learning on financial services, compliances and supervision is discussed. Qualitative and analytical approach is used for studying of financial regulations, different case studies and industry reports which helps in revealing that AI helps in improving the compliance checks, fraud detection as well as risk assessment.

Rajeshwari M. et. al., 2022 develop an AI-driven financial management system by making use of parallel computing for enhancing the decision making, risk prediction and increasing the efficiency of SMEs. Integration of AI algorithm is done by using computational and simulation based approach for testing the performance of system for processing of financial data, risk assessment and automation by means of hypothetical scenarios. The results reveals that integration of AI and parallel computing helps in improving accuracy, reduction in processing time along with streamlining the budget and investment plans.

BIG DATA ANALYTICS APPROACH FOR FINANCIAL MANAGEMENT:

S. Kumar et. al., 2022 focus on sustainable finance by making use of data analytics and machine learning. Paper focus on seven major themes as climate finance, social responsible investigation, impact investing, green finance, carbon finance, energy finance and governance of sustainable finance and investing. This paper proposed framework along with unifying and devising policies for sustainable finance. The paper also discusses the impact of new technologies like artificial intelligence, block chain, machine learning and their contribution to sustainable finance. For this approach, methods for collection of data, assembling of data and arranging of data along with assessing of data is explained which helps in performance analysis and science mapping for sustainable finance.

Hasan M. et. al, 2020 make use of big data technology in internet credit service-companies, finance markets along with internet finance, finance management, their analysis and applications, credit bank risk analysis, risk management etc. This is achieve by dividing area into three groups as big data implications for financial markets, big data implications in internet finance along with value creation in internet credit-service companies and third as big data in risk management, financial management, analysis of financial data and applications. Thus, this paper shows the pivotal role of big data in all three financial sectors.

uses big data in financial markets, internet credit service-companies and internet finance, financial management, analysis and applications, credit banking risk analysis, risk management etc. In this approach these areas are divided into three groups firstly big data implications for financial markets, secondly big data implications for internet finance and value creation in internet credit-service companies and thirdly big data in financial management, risk management, financial analysis, and applications. It concludes that big data plays a pivotal role in all these three financial sectors.

TABLE 1: COMPARATIVE ANALYSIS OF FINANCIAL BUDGETING

S. No.	Title of Paper	Tools and Technologies Used	Work Done	Result
1.	Financial development and economic	Pedroni Panel Cointegration Test,	Understand the relationship between credit	Credit-output growth for BIMAARU

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	growth in Indian States: an examination	FMOLS, VAR's	growth and output growth	states is lower
2.	The use of predictive analytics in finance	classification, regression, clustering, association and time series models	Understand the relationship between predictive analytics and finance	Effect of predictive analysis in finance
3.	Past, present, and future of sustainable finance: insights from big data analytics through machine learning of scholarly research.	Collection, assembling, arranging and assessing of data on sustainable finance	Understand seven major themes of sustainable finance	Performance analysis and science mapping for sustainable finance
4.	An exploratory data analysis approach for analyzing financial accounting data using machine learning	Linear Regression, K-Nearest Neighbor, Support Vector Regressor, and Decision Tree, grid search	Analyze financial accounting data, balance sheets, income statements, and cash flow statement.	Decision tree model results in optimized performance
5.	Finance for the Environment: A Scientometrics Analysis of Green Finance	Survey paper	Journals and papers related to green finance were discussed	Journals and papers related to green finance were discussed
6.	Construction of Sci-Tech Financial Ecosystem Indicators and International Comparative Research	Index system of Scitech (designing, data processing)	Sci-Tech framework is discussed and comparison of china with other countries is studied	Sci-tech framework developed
7.	Big data analytics for financial Market volatility forecast based on support vector machine	hybrid theory and support vector machine (SVM) algorithm	extract the characteristics of financial market, and analyze the relevant data of financial market.	Volatility is an important measurement index of market risk, Prediction of jump volatility and short-term volatility of high-frequency data is done.
8.	Current landscape and influence of big data on finance	Big Data	Analyze the use of big data in managing the financial data	Influence of big data on almost every aspect of financial markets, financial institutionsand so on.
9.	AI in Dynamic Budgeting and Forecasting	AI (machine learning, neural networks, and predictive analytic)	ML is used for analysis of historical and real-time data, enabling the creation of models that can predict future financial scenarios.	AI-driven budgeting models for enhanced dynamic budgeting and forecasting.
10.	Integrating AI Techniques for Enhanced Financial Forecasting and Budgeting Strategies	AI	Discusses uses Artificial Intelligence (AI) for Financial Forecasting and Budgeting for business decision-making	AI helps financial planners to concentrate on strategic analysis, aligning financial strategies seamlessly with organizational objectives.
11.	Machine learning for financial forecasting, planning and analysis: recent development and pitfalls	ML	machine learning for forecasting, planning and analysis of financial data.	Accuracy of sales forecasting and effectiveness of promotional activities achieved.
12.	Framework of the Smart Finance and Accounting Management Model	AI	proposed three AI driven algorithm as Earliest Deadline First (EDF), Highest Value First (HVF), and Highest	enhancing efficiency, financial process optimization along with support for real-time decision making

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	under the Artificial Intelligence Perspective.		Value Density First (HVDF)	
13.	Directed credit, financial development and financial structure: theory and evidence	theoretical model and empirical analysis	examining the directed credit policies on financial development, financial liberalization, firm financing and economic growth.	credit program can support financial development and economic growth under specific conditions.
14.	Optimal Analysis for Enterprise Financial Management Based on Artificial Intelligence and Parallel Computing Method	AI	develop an AI-driven financial management system by making use of parallel computing for enhancing the decision making, risk prediction and increasing the efficiency of SMEs.	AI and parallel computing improves accurac, reduction in processing time and streamlining the budget and investment plans.

Table 1 shows the representation of papers in the field of financial budgeting based on parameters like technology used, work done and results.

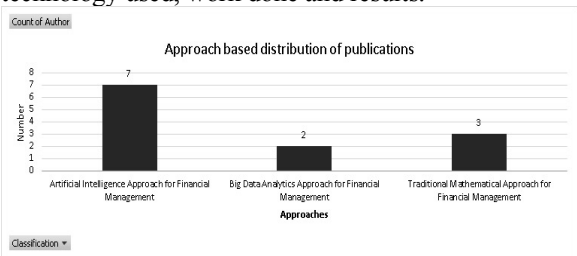

Fig 4: Graphical representation of Classification of financial budgeting

Figure 4 shows the graphical representation of classification of financial budgeting which represents that most of the paper of financial budgeting are based on artificial intelligence (i.e 7) which is highest whereas traditional methods approach are at second (i.e 3) whereas the lowest is that of big data (i.e 2) hence big data is not much appreciable approach towards financial budgeting whereas artificial intelligence based approach is considered to be the best approach for financial budgeting.

IV. CONCLUSION

The paper makes a review analysis of financial and budgeting analysis using mathematical approach, AI based approach and big data based approach. Traditional financial budgeting methods have inherent limitations in handling the complexities of modern financial data. Computational intelligence, incorporating AI, big data analytics, and advanced mathematical modeling, offers promising solutions for improving financial decision-making and optimizing resource utilization. The reviewed studies indicate that while AI-driven financial analytics is widely used globally, it has not been extensively implemented in India, particularly in Uttarakhand. The integration of these advanced techniques can help the state make more informed financial decisions, ultimately leading to better economic outcomes and efficient resource management. The study recommends that future research should focus on the implementation of AI-based financial models tailored to the unique economic conditions of Uttarakhand.

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